

Modality-Projection Universal Model for Comprehensive Full-Body Medical Imaging Segmentation

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

In their paper "Modality-Projection Universal Model for Comprehensive Full-Body Medical Imaging Segmentation", the authors develop a novel projection strategy to efficiently leverage multimodal representations to train a whole-body segmentation network. Core to their idea is learning a modality-agnostic latent feature vector T . To model the relationship between T and the "observed" latent representations (from each imaging modality), the authors employ representations from large (foundation?) models (their average) as "surrogate T ". Comparing their model to several baselines, they report outperformance for (1) segmentation, (2) detection assistance and (3) uncovering systemic correlations. While the work is interesting, I have key issues with the paper, in particular the reproducibility / detailed descriptions. Also, use case 3 (and, to a lesser extent, 2) fall short in highlighting the model's value and could be improved, as outlined below.

Major:

Case 2 (Sec. 2.2) -> Was the performance increase driven mostly by detecting hemorrhage at all (which would be most relevant) or correctly localizing it, as the text in Sec. 2.2 suggests? The latter would be of less clinical relevance.

Case 3: While I see the value of highlighting how anatomy segmentation can aid detection of region-spanning alterations, I am not convinced of the quality of this specific use case. First, the age range for the epilepsy patients is wide (2 - 18 yrs). Also, no detailed information on the type and severity of epilepsy is given. Is the control cohort age-matched (probably not, or why would you do a whole-body PET scan on a healthy 2yr old)? If not, then many changes between cohorts could also be attributed to the age difference? Also, is there any adjustment for multiple testing or are these spurious correlations?

Please clarify the methodology, in particular in view of the learned ("controller-based") convolution layer. How exactly is the output of the MLP shaped - a 3x3x3 kernel?

Many detailed information on the training (including the loss function), as well as for the competitor methods are missing, precluding a thorough evaluation. For example, how are the "parameters adjusted to match the complexity of the other models" for the "classic UNet"?

Core to your work is the "estimated T " from "external pre-trained models as additional projections of the latent features." (Sec 4.4). However, nowhere are these models detailed? CLIP, for example, is not a model, but a contrastive training strategy.

Along this line, one (or several) of these baseline models would be a strong competition; i.e., finetune an image foundation model to case 2

Minor:

> Discussion, sec. 3.1, line 621: "As shown in Case 3" -> Should be Case 2?

> Discussion, sec. 3.2, lines 636-639: Your citation [19] you use to support the investigation of the brain-body axis in epilepsy does not talk about epilepsy at all?

> Methods, 4.3.1: "For MR patches, the voxel values were divided by 3000 for normalization" -> Since MRI is a qualitative measurement, this normalization by an absolute value is uncommon. Why not also scale to [0;1]?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

This work targets the challenges of training multi-modal medical images from CT/MRI/PET, etc, specifically for segmentation tasks. The authors pose that training different modalities at the same time has big difficulties as organ features and modality features can be tangled together, resulting in training instability. To solve the challenges, this work proposed modality projection method as a universal model, the model can train and segment different organs across different modalities at the same time.

Pros:

1. The authors did a good review and investigation on the current challenges of training a universal model.
2. This paper is comprehensive and easy to follow.
3. This paper is written from the more clinical impact point of view, adding more value to a segmentation model paper. This is a rare paper that evaluates the diagnostic effect of performance with clinical studies.
4. The authors demonstrate great observations of body-body and body-brain associations, which are important quantitative studies from automated segmentation techniques.
5. The body-brain metabolic connection study is critical for discovering new biomarkers for many diseases.

Below are some major comments on concerns and issues:

1. Training a universal model across modalities is not a new topic. This paper proposed a good method, but it's hard to stand out from prior works on methodology.
2. Training a universal model from 861 subjects is not enough. Why not combine more public data for training and evaluate the effect of at least a pre-trained segmentation model? The capacity of training data is a major concern for a "universal model.", for example, the MedSAM work, which is mentioned in this paper's introduction, is trained over thousands of subjects.
3. In section 2.3, the authors show good application on using the universal segmentation tech for identifying association ROIs, pairs. From the description, after quantifying the ROIs, the identification of association pairs still needs massive manual efforts. Can this process be automated? Or use as ground truth for training the model, which can help segmentation tasks as well as pre-quantify the regions.
4. The body-brain pair for analyzing metabolic associations is a good effort. Is there any association or discovery of any disease-related biomarkers? Such as diabetes/alzheimer? This part can be extended with more description.
5. From the technical aspect, this paper lacks a general comparison of the proposed universal model; it needs to justify the effectiveness of the multi-modal and training strategy.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

Thank you for addressing the points raised. The manuscript has clearly improved as a consequence.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors addressed the reviewer's comments and concerns. The authors improved the quality of the manuscript with clearer images, equations, and discussions. The summarized Table 2 with detailed Dice scores comparison is a big milestone, well highlighting the advantage and assessment of the method.

As I have commented in the first round review, this paper has a unique impact on how to do translational research from a pioneering deep learning model to clinical settings. The authors can further add the discussion of the value of AI's impact on real clinical workflow, and highlight how to bridge frontier tech with medicine.

(Remarks on code availability)

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On Revision of NCOMMS-24-81749

“Modality-Projection Universal Model for Comprehensive Full-Body Medical Imaging Segmentation”

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June 26, 2025

Dear Editor and Reviewers,

Thank you very much for your valuable feedback and suggestions on our manuscript [Manuscript ID: NCOMMS-24-81749]. We greatly appreciate the time and effort you have invested in reviewing our work, and we are grateful for the constructive comments that have helped us improve the quality of our manuscript. We have carefully considered all the comments and have made revisions to address the concerns raised.

Reply to Reviewer #2’s Comments

Comment 1.1: [Case 2 (Sec. 2.2)] Was the performance increase driven mostly by detecting hemorrhage at all (which would be most relevant) or correctly localizing it, as the text in Sec. 2.2 suggests? The latter would be of less clinical relevance.

Reply: Thank you for your valuable comment. We first reviewed evidence showing that the exact location of an ICH critically influences treatment and prognosis, then redesigned our experiment to disentangle performance gains from hemorrhage detection versus gains from precise localization.

Recent studies highlight that precise ICH localization significantly influences management and outcomes. Brainstem hemorrhages carry the worst prognosis, as they are significantly associated with higher mortality[1]. Furthermore, patients with brainstem hemorrhages experience significantly worse health-related quality of life, particularly in the pain domain, compared to those with cerebellar hemorrhages[1]. Intraventricular extension with resulting hydrocephalus markedly worsens outcome: a meta-analysis reveals that patients with ICH, IVH, and hydrocephalus have substantially higher 30- and 90-day mortality rates compared to those with ICH alone[2]. Likewise, lobar (cortical) ICH is associated with higher early mortality than deep hemorrhages (26.7% vs. 16.5% in one cohort)[3]. Pooled analyses indicate early surgical evacuation of lobar ICH does not significantly improve functional outcome[4]. In contrast, basal ganglia or thalamic hemorrhages can benefit from surgery in selected cases: a large cohort study demonstrated that patients with good preoperative motor function who underwent hematoma evacuation had favorable neurological outcomes at 3 months[5]. Together, these findings highlight the significant prognostic and therapeutic implications of distinguishing between lobar, deep, cerebellar, brainstem, and

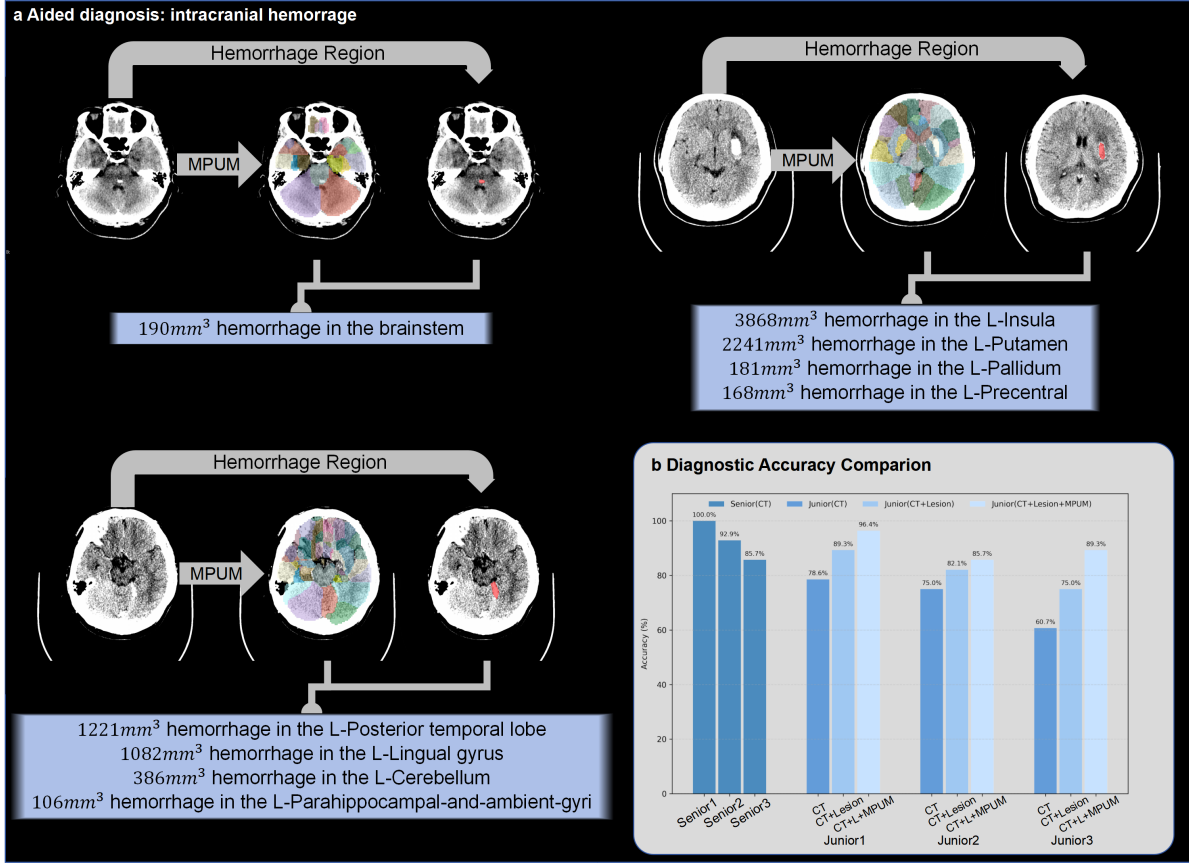


Figure 1: Performance of the MPUM Framework as an aided diagnosis tool. **a**, MPUM assists in hemorrhage detection and brain region mapping from CT head scans, facilitating accurate diagnosis. **b**, MPUM enhances diagnostic performance and supports non-specialist physicians in real-world clinical settings.

ventricular ICH.

To pinpoint the factors driving the performance gains and to further corroborate the model’s validity, we have adopted a more rigorous experimental design:

1. **CT only.** Both junior and senior radiologists reviewed raw CT volumes without any computer assistance. These results serve as baselines, with junior performance as the reference for measuring improvement and senior performance as the expert benchmark.
2. **CT + lesion mask.** Junior radiologists were provided with lesion masks generated by Viola[6], the top-performing model from the 2022 ICH Segmentation Challenge. Comparing this setting with the CT-only baseline quantifies the benefit of automated hemorrhage detection.
3. **CT + lesion mask + MPUM brain atlas.** Junior radiologists received both the lesion mask and the MPUM-derived brain region map. This setting evaluates whether providing anatomical localization in addition to lesion detection further improves diagnostic accuracy by assisting spatial reasoning and regional interpretation.

As summarised in Fig.1, without any assistance, senior radiologists achieved accuracies of 100%, 92.9%, and 85.7%, whereas junior radiologists reached only 78.6%, 75.0%, and 60.7%. Most junior radiologists’

errors stemmed from incomplete identification of all hemorrhage regions and confusion between brain regions. Supplying an automated lesion mask raised junior performance (e.g., Junior-1 from 78.6% to 89.3%, with similar gains for Junior-2 and Junior-3). When the MPUM brain-region atlas was also provided, their accuracies climbed further to 96.4%, 85.7%, and 89.3%, underscoring the utility of precise regional localization and the diagnostic value of the MPUM framework.

The automated lesion mask directs radiologists to voxels with high hemorrhage probability, shortening search time and minimizing the risk of missing small or low-contrast foci, thereby reducing false-negatives. The MPUM atlas provides a detailed anatomical map that clearly delineates boundaries, such as the interface between the temporal lobe and the lateral ventricle. Junior doctors often struggle to identify these boundaries accurately. By linking each highlighted lesion voxel to a specific neuroanatomical region, the atlas reduces region-of-interest confusion (e.g. mistaking parietal hemorrhage for ventricular bleeding), thereby boosting localization accuracy and specificity. In short, the lesion mask addresses the question “*Where is the bleed?*”, while the atlas answers “*Where, precisely, in neuro-anatomical terms?*”. Their combination replicates the workflow of expert radiologists and explains the stepwise performance gains observed in the study.

Comment 1.2: Case 3: While I see the value of highlighting how anatomy segmentation can aid detection of region-spanning alterations, I am not convinced of the quality of this specific use case. First, the age range for the epilepsy patients is wide (2 - 18 yrs). Also, no detailed information on the type and severity of epilepsy is given. Is the control cohort age-matched (probably not, or why would you a whole-body PET scan on a healthy 2yr old)? If not, then many changes between cohorts could also be attributed to the age difference?

Table 1: Demographic and clinical characteristics of the study cohorts

Variable	Level / Format	Epilepsy (n=50)	Control (n=22)
Sex	Male	23	17
	Female	27	5
Age(yrs)	Median (IQR)	12 (7, 15)	13 (7, 14)
Height(cm)	Median (IQR)	153 (134, 165)	156 (125, 175)
Weight(kg)	Mean $\pm$ SD	46.2 $\pm$ 19.9	46.5 $\pm$ 22.0
Seizure duration(yrs)	Median (IQR)	3 (2, 6)	–
Seizure frequency	Daily	14 (28 %)	–
	Weekly	8 (16 %)	–
	Monthly	13 (26 %)	–
	Yearly	5 (10 %)	–
	Irregular	10 (20 %)	–
Seizure type	Focal seizure	39 (78 %)	–
	Generalized seizure	11 (22 %)	–

Reply: We appreciate the reviewer’s thoughtful concerns and have addressed each point as follows.

Age matching and distribution. Detailed demographic statistics for both cohorts are now provided in Table 1. Median ages do not differ significantly between groups (Epilepsy: 12 years [IQR 7–15] vs. Control: 13 years [IQR 7–14]; Mann–Whitney U , $p = 0.782$). Height ($p = 0.586$) and weight ($p = 0.959$)

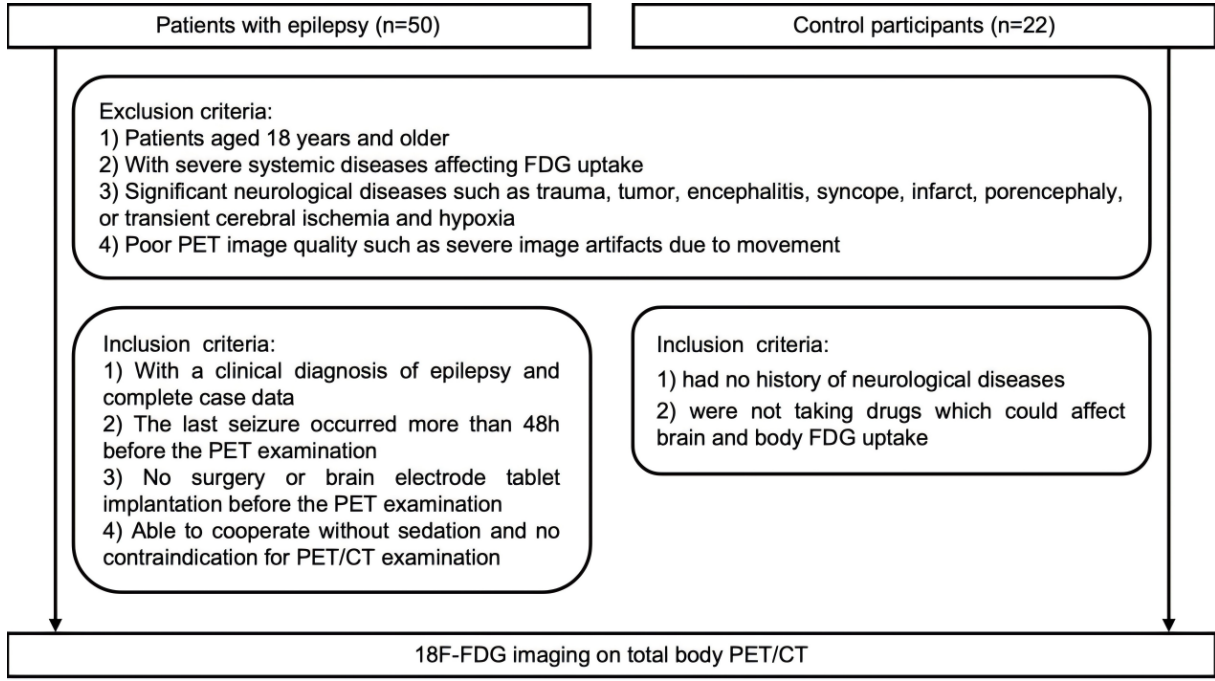


Figure 2: Flow-chart summarising exclusion and inclusion procedures for patients with epilepsy and control participants.

were likewise comparable, indicating an age- and size-matched control cohort.

Rationale for scanning very young patients. Whole-body 18F-FDG PET/CT was clinically indicated in the single 2-year-old case because the child had drug-resistant epilepsy starting at 7 months (2–3 seizures per day for 2 years). Metabolic imaging was required to localize the epileptogenic zone prior to surgical intervention.

Epilepsy type and severity. We now report seizure duration, frequency, and semiology for all 50 patients in Table 1. Specifically, 78% had focal seizures, and 22% had generalized seizures.

Inclusion and exclusion criteria. A flow-chart summarising screening, exclusion, and inclusion procedures for both cohorts has been added as Fig. 2.

Age-partial correlation analysis (method):

We applied the Fisher Z-transformation to compare correlation coefficients between two independent groups. This transformation standardizes raw correlation coefficients into values approximating a normal distribution, enabling hypothesis testing between control and patient groups.

In our analysis, we analyzed metabolic data from organs a and b in a control group (22 individuals), calculating the mean standardized uptake value (SUV) for the regions of interest (ROI) within these organs. The metabolic data for organ a in the control group is denoted as $A^{control} = \{a^n | n = 1, 2, \dots, 22\}$, representing the set of mean SUVs for each individual. Similarly, the metabolic data for organ b in the control group is represented as $B^{control} = \{b^n | n = 1, 2, \dots, 22\}$. We calculated the Pearson correlation coefficient, which measures the strength and direction of a linear relationship between two variables:

$$r_{a,b}^{group} = \frac{\sum_{i=1}^n (a^i - \bar{a})(b^i - \bar{b})}{\sqrt{\sum_{i=1}^n (a^i - \bar{a})^2 (b^i - \bar{b})^2}}, \quad group \in \{\text{control, patient}\}. \quad (1)$$

where a^i and b^i are the individual SUV for organs a and b in the control group, and $\bar{a}$ and $\bar{b}$ are the mean SUVs for organ a and b respectively. We can simplify the Eq.1 to $r_{a,b}^{group} = \text{pearson}(A^{group}, B^{group})$. Similarly, replacing an organ variable with Age yields the Pearson correlations $r_{a,Age}^{group}$ and $r_{b,Age}^{group}$.

To remove the linear effect of age, we calculated the partial correlation between organs a and b , conditional on age, using the closed-form solution:

$$r_{a,b|Age}^{group} = \frac{r_{a,b}^{group} - r_{a,Age}^{group} r_{b,Age}^{group}}{\sqrt{(1 - (r_{a,Age}^{group})^2)(1 - (r_{b,Age}^{group})^2)}}, \quad group \in \{\text{control, patient}\}. \quad (2)$$

Eq. 2 is equivalent to regressing a and b on age, extracting their residuals, and calculating the ordinary Pearson correlation between the residuals.

We then apply the Fisher Z-transformation to $r_{a,b}^{control}$ and $r_{a,b}^{patient}$:

$$z_{a,b|Age}^{control} = \text{arctanh}(r_{a,b|Age}^{control}) \quad (3)$$

$$z_{a,b|Age}^{patient} = \text{arctanh}(r_{a,b|Age}^{patient}) \quad (4)$$

And computed the Z-score:

$$z = \frac{z_{a,b|Age}^{control} - z_{a,b|Age}^{patient}}{\sqrt{1/(n^{control} - 3) + 1/(n^{patient} - 3)}}, \quad (5)$$

where $n^{control}$ and $n^{patient}$ are the sample sizes of control group and patient group, respectively. Finally, we calculate the p-value assuming a normal distribution:

$$p = 2 * (1 - \text{cdf}(\text{abs}(z))), \quad (6)$$

where $\text{abs}(z)$ represent the absolute value of the Z-score and cdf stands for cumulative distribution function. The term $1 - \text{cdf}(\text{abs}(z))$ represents the one-tailed p-value, and multiplying by 2 makes it a two-tailed test.

Comment 1.3: Case 3: Also, is there any adjustment for multiple testing or are these spurious correlations?

Reply: We appreciate the reviewer's insightful comment.

In whole-body and multi-organ metabolic studies, approaches to multiple testing correction have varied. For instance, one work[7] examined 126 subjects (63 diabetes, 63 controls) across 11 organs using independent-sample t-tests and Pearson correlations without multiplicity correction. Conversely, another work[8] applied a Bonferroni correction to 153 lung-cancer network edges ($\alpha=0.01$). The work[9] combined Bonferroni correction for edge-wise tests ($\alpha=0.05$) with FDR ($\alpha=0.05$) for voxel-wise tests. Likewise, the study[10] controlled the FDR when comparing pMCI vs. sMCI metabolic connectomes.

a R Anterior temporal lobe lateral part				b R-Middle and inferior temporal gyrus			
BrainRegion	Corr	P-value	FDR P-value	BrainRegion	Corr	P-value	FDR P-value
L-Straight gyrus	0.9842 >>> 0.6355	4.7e-13	8.4e-10	L-Inferior frontal gyrus	0.9858 >>> 0.8226	3.4e-09	6.6e-07
R-Straight gyrus	0.9721 >>> 0.6452	1.2e-10	6.7e-08	L-Middle and inferior temporal gyrus	0.9794 >>> 0.8612	8.8e-09	1.4e-06
L-Medial orbital gyrus	0.9814 >>> 0.6877	7.6e-10	2.1e-07	R-Inferior lateral remainder of parietal lobe	0.9903 >>> 0.9548	3.9e-08	4.3e-06
L-Superior frontal gyrus	0.9805 >>> 0.6608	8.2e-09	1.3e-06	L-Straight gyrus	0.9859 >>> 0.7916	1.1e-07	9.6e-06
R-Inferior lateral remainder of parietal lobe	0.9765 >>> 0.7511	8.3e-09	1.3e-06	L-Superior frontal gyrus	0.9780 >>> 0.8320	3.5e-07	2.2e-05
L-Inferior frontal gyrus	0.9704 >>> 0.7381	1.1e-08	1.6e-06	R-Straight gyrus	0.9692 >>> 0.7965	5.9e-07	3.5e-05
L-Insula	0.9593 >>> 0.6597	4.0e-08	4.4e-06	R-Postcentral gyrus	0.9687 >>> 0.8349	7.7e-07	4.2e-05
R-Superior frontal gyrus	0.9742 >>> 0.6928	4.2e-08	4.6e-06	L-Medial orbital gyrus	0.9809 >>> 0.8338	8.3e-07	4.4e-05
L-Anterior orbital gyrus	0.9562 >>> 0.5899	5.4e-08	5.6e-06	R-Posterior temporal lobe	0.9791 >>> 0.9235	1.1e-06	5.4e-05
R-Middle frontal gyrus	0.9802 >>> 0.7265	6.9e-08	6.8e-06	R-Inferior frontal gyrus	0.9880 >>> 0.8713	1.3e-06	6.2e-05
R-Medial orbital gyrus	0.9789 >>> 0.7302	8.3e-08	7.8e-06	L-Middle frontal gyrus	0.9836 >>> 0.8425	1.8e-06	7.9e-05
R-Posterior orbital gyrus	0.9729 >>> 0.7772	1.3e-07	1.0e-05	R-Superior frontal gyrus	0.9728 >>> 0.8312	3.6e-06	1.3e-04
R-Postcentral gyrus	0.9795 >>> 0.7105	2.0e-07	1.5e-05	L-Anterior orbital gyrus	0.9769 >>> 0.7535	4.1e-06	1.5e-04
L-Middle frontal gyrus	0.9591 >>> 0.6808	2.1e-07	1.5e-05	L-Posterior temporal lobe	0.9695 >>> 0.8542	5.5e-06	1.9e-04
L-Middle and inferior temporal gyrus	0.9801 >>> 0.7574	2.8e-07	1.8e-05	R-Medial orbital gyrus	0.9832 >>> 0.8476	7.2e-06	2.4e-04
L-Fusiform gyrus	0.9718 >>> 0.6774	2.9e-07	1.9e-05	L-Superior parietal gyrus	0.9779 >>> 0.8275	1.8e-05	4.7e-04
R-Posterior temporal lobe	0.9752 >>> 0.6001	6.7e-07	3.8e-05	R-Middle frontal gyrus	0.9780 >>> 0.8821	2.1e-05	5.2e-04
R-Inferior frontal gyrus	0.9731 >>> 0.6091	7.5e-07	4.1e-05	L-Lateral orbital gyrus	0.9574 >>> 0.7509	3.5e-05	7.6e-04
R-Anterior orbital gyrus	0.9720 >>> 0.6987	8.1e-07	4.3e-05	R-Posterior orbital gyrus	0.9855 >>> 0.8929	3.7e-05	8.0e-04
L-Posterior orbital gyrus	0.9677 >>> 0.6496	8.2e-07	4.3e-05	L-Anterior-temporal-lobe-lateral-part	0.9756 >>> 0.8113	3.8e-05	8.1e-04
L-Superior-temporal-gyrus-posterior-part	0.9716 >>> 0.7117	9.0e-07	4.6e-05	R-Lateral orbital gyrus	0.9691 >>> 0.7770	3.8e-05	8.1e-04
L-Superior parietal gyrus	0.9515 >>> 0.6333	1.4e-06	6.4e-05	L-Precentral gyrus	0.9669 >>> 0.8449	5.2e-05	1.0e-03
L-Precentral gyrus	0.9569 >>> 0.7148	1.7e-06	7.4e-05	L-Fusiform gyrus	0.9606 >>> 0.8221	5.7e-05	1.1e-03
L-Posterior temporal lobe	0.9601 >>> 0.7447	1.8e-06	7.9e-05	R-Anterior-temporal-lobe-lateral-part	0.9708 >>> 0.8993	7.9e-05	1.4e-03
L-Postcentral gyrus	0.9629 >>> 0.6820	3.0e-06	1.2e-04	L-Superior temporal gyrus anterior part	0.9673 >>> 0.8011	9.4e-05	1.6e-03
R-Insula	0.9517 >>> 0.6525	6.0e-06	2.1e-04	L-Superior-temporal-gyrus-posterior-part	0.9653 >>> 0.8263	1.2e-04	1.9e-03
L-Lateral orbital gyrus	0.9537 >>> 0.7021	6.3e-06	2.1e-04	R-Lateral remainder of occipital lobe	0.9682 >>> 0.8757	1.3e-04	1.9e-03
L-Cerebellum	0.9573 >>> 0.6742	9.7e-06	3.0e-04	L-Postcentral gyrus	0.9706 >>> 0.8386	1.8e-04	2.5e-03
R-Cerebellum	0.9460 >>> 0.6679	1.0e-05	3.0e-04	L-Lateral remainder of occipital lobe	0.9585 >>> 0.8418	3.5e-04	4.2e-03
L-Superior temporal gyrus anterior part	0.9499 >>> 0.6027	1.1e-05	3.2e-04	L-Insula	0.9574 >>> 0.8313	4.0e-04	4.6e-03
L-Anterior-temporal-lobe-medial-part	0.9706 >>> 0.7352	1.6e-05	4.4e-04	L-Anterior-temporal-lobe-medial-part	0.9606 >>> 0.8070	4.8e-04	5.2e-03
R-Lateral orbital gyrus	0.9681 >>> 0.7877	1.7e-05	4.5e-04	R-Cerebellum	0.9570 >>> 0.7990	5.6e-04	6.0e-03
R-Superior temporal gyrus anterior part	0.9777 >>> 0.8202	1.8e-05	4.6e-04	L-Posterior orbital gyrus	0.9594 >>> 0.7876	6.8e-04	6.9e-03
R-Superior-temporal-gyrus-posterior-part	0.9604 >>> 0.7741	2.2e-05	5.5e-04	L-Cerebellum	0.9487 >>> 0.7856	7.6e-04	7.5e-03
R-Superior parietal gyrus	0.9513 >>> 0.7006	2.4e-05	5.8e-04	R-Superior parietal gyrus	0.9573 >>> 0.8328	1.1e-03	9.9e-03
R-Precentral gyrus	0.9482 >>> 0.7098	2.5e-05	6.0e-04				
R-Lateral remainder of occipital lobe	0.9672 >>> 0.7342	2.7e-05	6.4e-04				
L-Anterior-temporal-lobe-lateral-part	0.9587 >>> 0.7217	4.3e-05	8.8e-04				
R-Subgenual frontal cortex	0.9158 >>> 0.4736	7.8e-05	1.4e-03				
R-Middle and inferior temporal gyrus	0.9743 >>> 0.8987	7.9e-05	1.4e-03				
L-Parahippocampal-and-ambient-gyri	0.9533 >>> 0.7254	1.2e-04	1.8e-03				
L-Subcallosal area	0.9079 >>> 0.4649	1.2e-04	1.8e-03				
L-Lateral remainder of occipital lobe	0.9441 >>> 0.7113	2.2e-04	2.9e-03				
R-Subcallosal area	0.9204 >>> 0.5282	2.4e-04	3.1e-03				
L-Cingulate gyrus gyrus cinguli posterior part	0.9327 >>> 0.6882	2.7e-04	3.4e-03				
L-Lingual gyrus	0.9246 >>> 0.6601	2.9e-04	3.6e-03				
L-Thalamus	0.9345 >>> 0.6392	3.3e-04	4.0e-03				
L-Caudate nucleus	0.9298 >>> 0.6938	3.6e-04	4.3e-03				
Brainstem	0.9406 >>> 0.6591	4.2e-04	4.7e-03				
R-Cingulate gyrus gyrus cinguli anterior part	0.9428 >>> 0.6920	5.9e-04	6.2e-03				
L-Cuneus	0.9133 >>> 0.6486	6.2e-04	6.4e-03				
R-Pallidum	0.8827 >>> 0.4744	7.5e-04	7.4e-03				
R-Thalamus	0.9412 >>> 0.7637	9.2e-04	8.7e-03				
L-Inferior lateral remainder of parietal lobe	0.9316 >>> 0.7310	9.5e-04	8.9e-03				

c Pallidum			
BrainRegion	Corr	P-value	FDR P-value
rib_left_9	0.7492 >>> 0.0115	2.9e-04	3.6e-03
rib_right_9	0.7197 >>> 0.0452	8.9e-04	8.5e-03
vertebrae_T1	0.7999 >>> 0.1117	1.7e-04	2.4e-03
vertebrae_T2	0.7705 >>> 0.1128	7.1e-04	7.1e-03
vertebrae_T3	0.8069 >>> 0.1049	1.0e-04	1.6e-03
vertebrae_T4	0.8364 >>> 0.1062	1.4e-05	3.9e-04
vertebrae_T5	0.7656 >>> 0.0922	3.2e-04	3.9e-03
vertebrae_T6	0.7974 >>> 0.0931	1.4e-04	2.0e-03
vertebrae_T7	0.8100 >>> 0.1117	6.5e-05	1.2e-03
vertebrae_T8	0.7783 >>> 0.0295	2.5e-04	3.2e-03
vertebrae_T9	0.7165 >>> 0.0288	1.3e-04	1.9e-03
vertebrae_T10	0.7526 >>> 0.1042	9.9e-04	9.2e-03
vertebrae_T11	0.7868 >>> 0.1435	9.6e-04	9.0e-03
vertebrae_T12	0.7462 >>> 0.1473	7.0e-04	7.1e-03

Figure 3: The impact of epilepsy on the metabolic correlations of organs. All results shown are statistically significant at $\alpha = 0.01$ with FDR correction. The “Corr” column in the table is formatted as “A>>>B,” where value A represents the metabolic correlation between two regions under the control group, and value B indicates that in the patient group. **a**, displays all brain regions related to the “R Anterior temporal lobe lateral part”. **b**, displays all brain regions related to the “right middle and inferior temporal gyrus”. **a&b** show inter-regional brain metabolic associations (brain-brain pairs). **c**, displays significant change in brain-to-body metabolic associations (brain-body pairs).

Controlling for multiple comparisons is critical in high-dimensional network analyses, where testing thousands of edges simultaneously (e.g., $> 20,000$ pairwise correlations in our study) dramatically increases the risk of false-positive findings. To address this, we rigorously evaluated two established approaches:

- 1. Bonferroni Correction** This method strictly controls the family-wise error rate (FWER) by dividing the nominal significance threshold ($\alpha = 0.01$ in our study) by the total number of tests. For our network analysis, this would require a Bonferroni-adjusted threshold of $0.01/20,000 = 510^{-7}$. While effective at minimizing type-I errors, this approach becomes prohibitively conservative in large-scale studies.
- 2. False Discovery Rate (FDR)** In contrast, the Benjamini-Hochberg FDR procedure controls the expected proportion of false discoveries among statistically significant results rather than the absolute probability of any false positive. As demonstrated in Fig. 3, we report both raw p-values and FDR-adjusted p-values for full transparency. Importantly, all results remain statistically significant at $\alpha = 0.01$ after FDR correction, reinforcing the robustness of our conclusions.

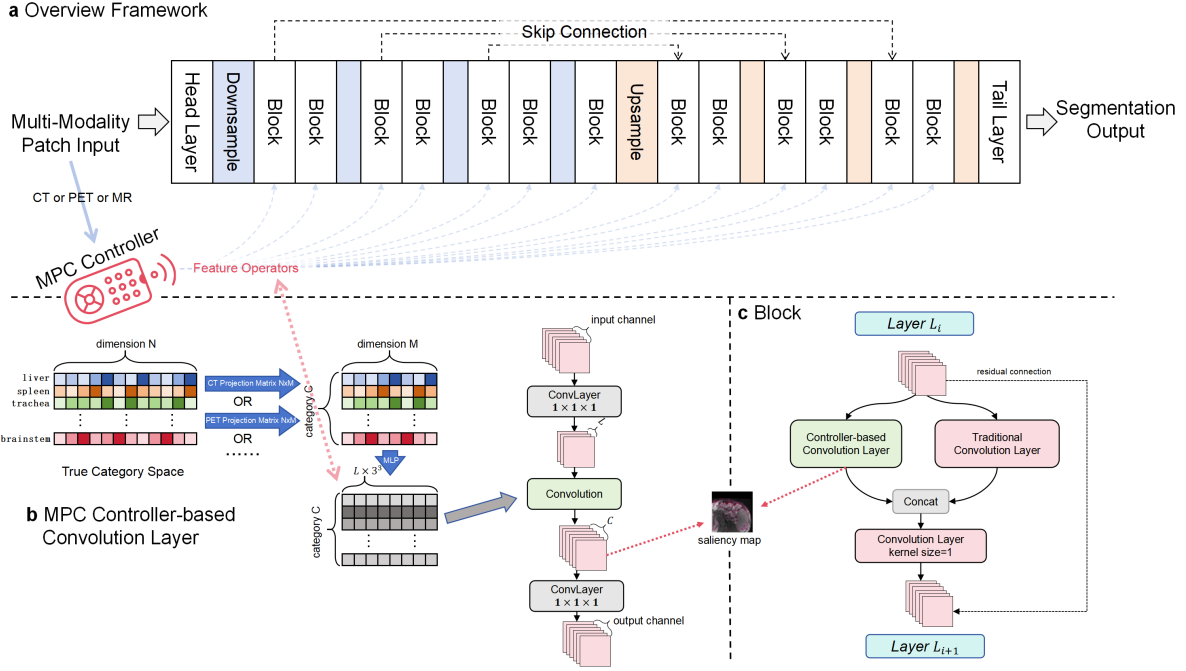


Figure 4: Overview and detailed structure of the MPUM. **a**, shows the overview framework of MPUM. The modality-projection controller influences the feature operators within each block based on the input modality. **b**, provides a detailed mechanism of the controller. Each input category has a latent feature with dimension N . Through different modality projection matrices, various modality feature spaces with dimension M can be obtained. Subsequently, these features are transformed into feature operators via a multi-layer perceptron. **c**, displays the detailed structure of the block. Each block contains two branches: one is a traditional convolutional layer, and the other is a non-parametric controller-based convolution layer, whose kernel is generated by the controller. The output feature maps from the two branches are merged through concatenation. The output feature map of the controller-based convolution layer also serves as the saliency map, providing interpretability.

Comment 1.4: Please clarify the methodology, in particular in view of the learned (“controller-based”) convolution layer. How exactly is the output of the MLP shaped - a $3 \times 3 \times 3$ kernel?

Reply: Thank you for raising this comment regarding the methodology. We clarify the design and implementation of the controller-based convolution layer.

The controller module dynamically generates 3D convolution kernels conditioned on modality-specific embeddings, enabling adaptive feature extraction across multi-modal medical imaging data. This replaces traditional static kernels with learnable, input-dependent parameters.

As illustrated in Fig. 4, we detail the operation of the controller-based convolution layer and clarify how the MLP output is shaped into a 3D kernel. Firstly, the input feature map is projected into a latent space via a $1 \times 1 \times 1$ convolution, reducing its channel dimension to a predefined latent size L (e.g. $L = 64$). This step compresses the feature map into a latent representation while preserving spatial dimensions. The multi-layer perceptron (MLP) in the controller then outputs a flattened kernel of length $L \times 3^3$, where 3^3 corresponds to the spatial dimensions of the kernel. Finally, the filtered feature map

is projected back to the target output channel dimension using another $1 \times 1 \times 1$ convolution, ensuring compatibility with downstream operations.

The MPC uses an MLP to generate convolution kernel parameters, a methodology widely validated in prior work. For instance, HyperNetworks[11, 12] pioneered the use of auxiliary networks to dynamically generate weights for primary network, demonstrating the feasibility of MLP-based parameter synthesis. Similarly, HyperConvolution[13] extended this concept by leveraging MLPs to produce continuous convolution kernels. While our approach adopts this mechanism, our innovation lies in the use of modality-specific embeddings to guide kernel generation, encoding anatomical priors specific to multi-modal medical data.

Comment 1.5: Many detailed information on the training (including the loss function), as well as for the competitor methods are missing, precluding a thorough evaluation. For example, how are the “parameters adjusted to match the complexity of the other models” for the “classic UNet”?

Reply: We thank the reviewer for pointing out the need for greater transparency regarding training details and competitor implementations.

To ensure a fair and comprehensive comparison, we applied a consistent training protocol across all models. Specifically, we adopted standardized preprocessing (e.g., resampling to 2 mm isotropic resolution), patch-based training with $128 \times 128 \times 128$ input size, and augmentation strategies such as random Gaussian smoothing and contrast adjustment, as detailed in the Implementation section of the original manuscript. In addition, all networks were optimized with the Adam optimizer (initial learning rate $3e-4$, weight decay $3e-5$), with the learning rate reduced by a factor of 10 whenever the validation loss plateaued. We used a batch size of 4 and trained for up to 200 epochs, employing early stopping based on validation performance to avoid overfitting. The training objective was a combination of categorical cross-entropy and soft Dice loss. Each network’s weights were initialized with the He-normal method and trained from scratch.

Each competitor method was trained following its authors’ recommended procedure, using the same data splits and above-mentioned training hyperparameters. For STUNet, we employed the official nnU-Net-based training code and re-trained it from scratch on our dataset with augmentation and loss. For CDUM[14], we used the published CLIP-driven model. The image-text encoder was pre-trained and frozen, and we trained the segmentation network. For PCNet, we used the authors’ public repository, including the prior-category prompts they describe. In all cases, training used the same optimizer, learning-rate schedule, batch size, and stopping criteria. This allows a fair evaluation where any observed performance differences arise from architectural and algorithmic differences rather than from mismatched training protocols.

To eliminate network capacity as a confounding factor, we adjusted all models to have comparable numbers of trainable parameters. PCNet[15], MPUM, and STUNet[16] each have approximately 60 million trainable parameters, whereas the classic 3D U-Net contains around 19M. We scaled the classic 3D U-Net by increasing the number of feature channels at each stage by a factor of 1.75 (e.g., $(32 \rightarrow 56, 64 \rightarrow 112, \dots, 512 \rightarrow 896)$). This brought its parameter count to approximately 59M. By normalizing model capacity, we ensure that the performance differences genuinely reflect innovations in architecture and design, rather than differences in model complexity.

Comment 1.6: Core to your work is the “estimated T” from “external pre-trained models as addi-

tional projections of the latent features.” (Sec 4.4). However, nowhere are these models detailed? CLIP, for example, is not a model, but a contrastive training strategy.

Reply: We appreciate the reviewer’s insightful comment. We address the concerns in two parts: (1) clarifying the distinction between CLIP as a training strategy and pretrained model, and (2) providing a more detailed explanation of the external models employed in our framework.

CLIP[17] can refer to both a training paradigm and the resulting pretrained models:

- As a training strategy, CLIP is a contrastive learning framework that aligns paired image and text representations in a shared embedding space. It optimizes the similarity between matched image-text pairs while minimizing similarity between mismatches.
- As a pretrained model, OpenAI has publicly released specific implementations of CLIP, which are widely adopted as frozen feature extractors in downstream tasks[18, 19].

In our manuscript, when we refer to “CLIP”, we specifically mean the pretrained model, not the training strategy. We utilize CLIP’s frozen text encoder to embed semantic category prompts (e.g., “liver”, “frontal cortex”), generating vector representations used as auxiliary guidance to inform latent feature projections (denoted as T).

Comment 1.7: Along this line, one (or several) of these baseline models would be a strong competition; i.e., finetune an image foundation model to case 2 Add Case 1 as well.

Reply: We appreciate the reviewer’s insightful comment. While CLIP and BioCLIP models demonstrate effective feature-wise alignment between images and text through contrastive language-image pre-training, their application to precise segmentation tasks faces inherent limitations.

Classical contrastive learning methods[20, 21] focus on constructing positive/negative sample pairs via data augmentation to learn invariant feature representations. While this approach excels in image classification by maximizing global feature similarity between augmented views, it struggles with segmentation tasks requiring voxel-wise discriminative capabilities. Two critical limitations arise:

1. **Inter-subject bias:** Learned representations primarily capture inter-subject distinctions, which benefits disease classification/prognosis, but fail to preserve the spatial detail required for segmentation.
2. **Information compression:** Vision encoders output subject-level embeddings (e.g., a 1024-dimensional vector for an entire 3D volume). When cascaded with a segmentation decoder, this compressed representation is often too coarse to support precise voxel-wise decoding when passed through a segmentation head.

For example, a typical CT image in whole-body PET/CT scan ($948 \times 250 \times 250$ voxels at 1.5 mm^3 resolution) contains about 59 million voxels. Contrastive training would involve rotations/noise augmentations of the same scan as positives and other scans as negatives. While pixel-wise contrastive learning methods such as VADeR[22], MaskContrast[23], and STEGO[24] attempt to address these issues in 2D, extending them to 3D medical data presents major challenges. This introduces critical shortcomings for segmentation:

1. **Anatomic Inconsistency:** Spatial augmentations like flipping or cropping may disrupt physiologically structures in 3D volumes.

2. **Low Inter-class Feature Disparity:** Natural images typically exhibit clear boundaries and distinct color/texture variations between objects (e.g., sky vs. buildings), providing strong discriminative signals for contrastive learning. In CT scans, tissues are differentiated by subtle Hounsfield unit (HU) variations. In PET scan, metabolic activity boundaries are diffuse and unclear, thereby increase the difficulty of unsupervised contrastive learning.
3. **Computational Cost:** Pixel-wise negative pairs in 3D volumes induce prohibitive complexity. For example, a whole-body CT scan contains about 59 million voxels. Constructing negative pairs across all voxels (i.e., $\{p_i, p_j\}$ where $i \neq j$) would require 3.5×10^{15} pairs per scan, which is a heavy computation burden.

In summary, our framework leverages CLIP-style pre-trained models not through fine-tuning, but as lightweight, frozen priors (T) to guide latent feature projection. This avoids the computational overhead of full-scale contrastive training, while still enhancing anatomical semantic alignment. We agree that contrastive pretraining holds long-term promise for segmentation, and we will clarify these points and further discuss this direction as future work in the revised manuscript.

Comment 1.8: Discussion, sec. 3.1, line 621: "As shown in Case 3" - Should be Case 2?

Reply: Thanks for pointing this out. The reference was incorrect and should be "Case 2". We have corrected this in the revised manuscript.

Comment 1.9: Discussion, sec. 3.2, lines 636-639: Your citation [19] you use to support the investigation of the brain-body associations in epilepsy does not talk about epilepsy at all?

Reply: We sincerely appreciate the reviewer’s meticulous attention to citation accuracy. The citation to reference [19] in Sec. 3.2 was an unintentional error, as it should have referred to reference[25] (originally labeled as [40] in the initial manuscript). The corrected citation supports our argument that epilepsy may also involve systemic metabolic effects beyond the brain, with the following statement from[25]:

"Hypoglycemia is a common, acute life-threatening illness in diabetic patients that may cause seizures."

This evidence highlights how systemic glucose dysregulation triggers seizures through cerebral energy failure. While this example focuses on acute hypoglycemia, it illustrates a mechanistic link between peripheral metabolic conditions and seizure activity. We also revised the manuscript to incorporate two additional references[26, 27] that specifically investigate the brain–body axis in epilepsy.

Comment 1.10: Methods, 4.3.1: "For MR patches, the voxel values were divided by 3000 for normalization" → Since MRI is a qualitative measurement, this normalization by an absolute value is uncommon. Why not also scale to [0-1]?

Reply: Thank you for raising this critical point. Normalization by a fixed absolute value (e.g., dividing by 3000) is inappropriate for MRI, given its qualitative and non-standardized intensity scale. We apologize for the outdated and incorrect description in the manuscript. While our actual implementation used a percentile-based normalization approach, this was inadvertently not reflected in the manuscript text due to an oversight. Specifically, we applied intensity truncation at the 0.5-th and 99.5-th percentiles, followed by linear scaling to the [0, 1] range. All experimental results presented in the manuscript are based on models trained with this corrected normalization pipeline.

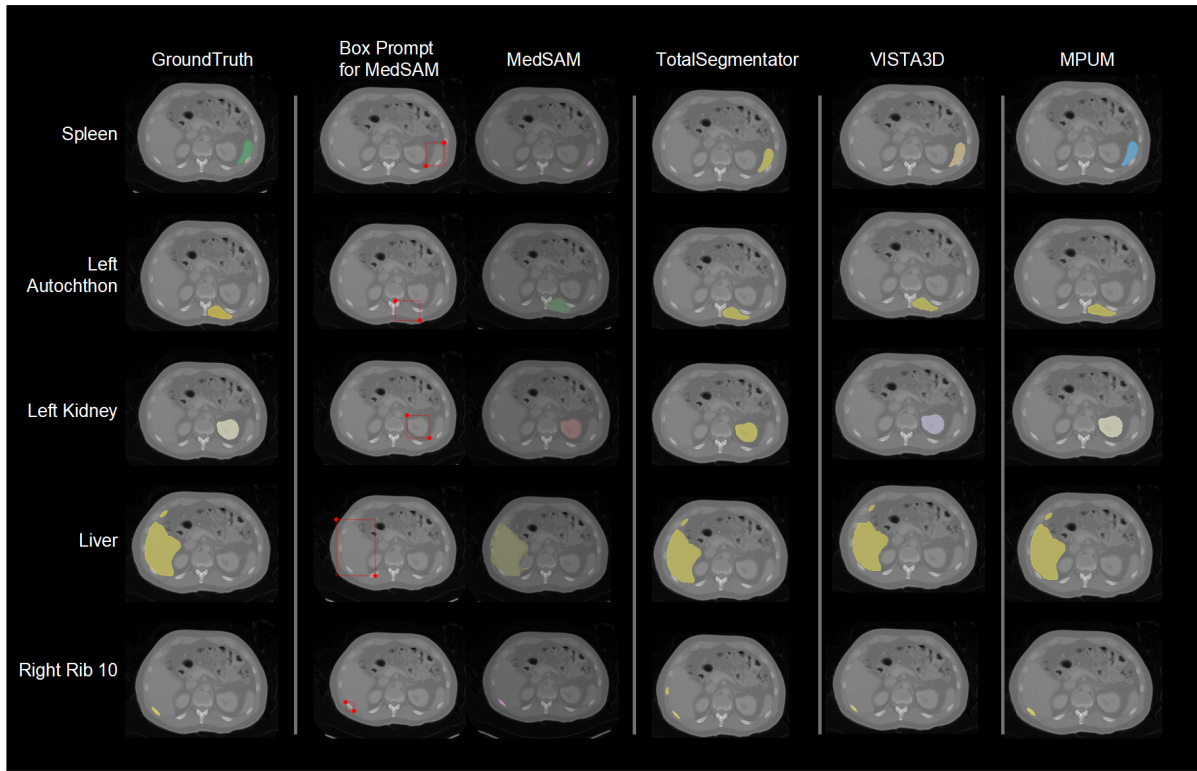


Figure 5: Visualization of segmentation results from multiple pre-trained models on the same CT case. The MedSAM output was generated by manually supplying box prompts.

We have made revisions in the manuscript accordingly, changing the description from “the voxel values were divided by 3000 for normalization” to “MR images were first intensity-truncated at the 0.5th and 99.5th percentiles to suppress outliers, then linearly scaled to the $[0, 1]$ range”. We apologize again for our oversight.

Reply to Reviewer #3’s Comments

Comment 2.1: Training a universal model across modalities is not a new topic. This paper proposed a good method, but it’s hard to stand out from prior works on methodology. Add more foundation/universal model.

Reply: We thank the reviewer for raising this question. There have already been several efforts toward universal cross-modal models. We provide an comprehensive overview of these works to highlight how our approach differs in its design philosophy and integration strategy.

Research on cross-modal foundation/universal segmentation has largely followed four paradigms:

1. **Prompt-driven Models:** These leverage explicit (points, boxes, text) to steer a single large model toward masks (e.g., MedSAM[28], UniSeg[29]). By fine-tuning the original Segment-Anything Model (SAM)[30] architecture on 1.57M medical image—mask pairs spanning 10 modalities, MedSAM[28] becomes the first prompt-driven foundation model able to segment 86 internal and 60 external medical tasks with the same point/box prompts used in SAM. The SAM-series models are interactive segmentation models that require specific prompts such as boxes or points to perform segmentation.

This limits their applicability in fully automated processing of large-scale data, such as case 3 in our manuscript. Moreover, their segmentation performance under certain modalities is suboptimal. As shown in Fig.5, although MedSAM accurately segments some structures (e.g., left kidney, right rib 10), it underperforms on others (e.g., liver, spleen) due to its reliance on prompt accuracy. For the spleen, its irregular shape forced the box prompt to include adjacent rib, causing MedSAM to mis-segment the rib and fail to isolate the spleen. For these reasons, we did not include SAM-based models in our automated benchmark.

2. **Structure-Adaptive Models:** These dynamically reconfigure network to handle varying modalities, or domain distributions (e.g., SPADNets[31], UniSeg[29]). SPADNet[31] is a 3D foundation framework whose convolutions automatically shrink or expand along the depth axis according to the voxel spacing of the incoming scan, so the same backbone can handle both 3D volumes and single-slice 2D images. UniSeg[29] keeps an nnUNet-style encoder-decoder but appends a learnable universal prompt P_u to the feature map before the decoder start. A lightweight gating MPL converts this prompt into weights that linearly mix 11 fixed $3 \times 3 \times 3$ expert kernels (one per training task) inside a single dynamic-convolution layer at the entrance of the decoder. The mixed kernel makes the decoder task-aware and lets one network handle 11 3D medical datasets. While elegant, UniSeg presents several limitations: (1) prompt is injected only after encoding, leaving early encoding layers blind to the current modality. Therefore, modality-specific knowledge is hard to be extracted by the encoders with same filters across different modalities. (2) UniSeg is elegant for switching among modalities, but it treats each modality in isolation. (3) The released UniSeg are trained to segment narrow anatomical scope with seven structures (liver, kidney, hepatic vessels, pancreas, spleen, vertebrae and prostate), limiting generalizability across categories.
3. **Native 3D Models:** These models operate directly on 3D voxels (e.g., VISTA3D[32], TotalSegmentator[33]). TotalSegmentator[33] finetunes an nnU-Net ensemble on whole-body CT exams, segmenting 104 anatomical structures spanning organs, bones, muscles, and vessels. TotalSegmentator achieves minute-scale full-body segmentation on standard GPUs. VISTA3D[32] contains a 3D UNET backbone with super-voxel distillation that injects pretrained 2D ViT-H features, letting the model learn full-volume context without GPU-prohibitive memory. It delivers state-of-the-art automatic segmentation for 127 classes.
4. **Few-shot/Zero-shot Models:** These focuses on enabling the model to infer accurately on new anatomies, modalities, or label sets with minimal additional annotations (e.g., UniverSeg[34]). UniverSeg[34] trains a single segmentation network on many anatomies and modalities and, at inference, feeds it a small “support set” of image-mask pairs to segment unseen region. A lightweight Cross-Block module propagates information from this support set through the backbone, so the model can segment the query image without any retraining or fine-tuning, achieving strong few-shot performance on completely unseen anatomies, modalities, and label sets. However, UniverSeg is a 2D model that requires support sets constructed from binary masks. This poses challenges for practical deployment. For example, when applying it to 3D whole-body scans, each volume must be sliced into 2D images, potentially generating thousands of support examples for a single category. This contradicts the model’s original goal of leveraging minimal support to segment unseen regions. Moreover, because UniverSeg does not optimize for any specific dataset, its performance on specific dataset remains inferior to fully supervised models. This represents a trade-off between generalisation and performance on specific tasks.

Our approach departs from these paradigms. We achieve latent-level integration by projecting modality-specific features from CT, MR, and PET into a shared high-dimensional space T , enabling mutual reinforcement and shared semantics across modalities, while also being fully automated without user prompts

and capable of category-level generalization across dozens of anatomical structures. We clarify this comparative analysis and the distinctions in the revised manuscript.

Comment 2.2: Training a universal model from 861 subjects is not enough. Why not combine more public data for training and evaluate the effect of at least a pre-trained segmentation model? The capacity of training data is a major concern for a “universal model.”, for example, the MedSAM work, which is mentioned in this paper’s introduction, is trained over thousands of subjects.

Reply: We sincerely thank the reviewer for raising this critical point. We fully agree that the capacity and diversity of training data are vital for developing universal segmentation models. Large-scale works like MedSAM[28] represent a significant milestone in this direction. However, we would like to clarify several important distinctions between our dataset and large-scale, partially annotated datasets like MedSAM, and explain why our data (smaller in subject count) offers high informational density that supports robust model training.

1. **High Annotation Density Compensates for Subject Volume.** While our dataset includes 861 subjects (533 CT, 533 PET, 328 MR), each CT scan is annotated with 214 structures, resulting in over 200,000 image-mask pairs across modalities. In contrast, many large-scale public datasets rely on partial, coarse, or prompt-derived annotations (e.g., MedSAM’s reliance on semi-automatic label generation). Fig. 6 illustrates that even with 30% of the dataset, our model achieves strong segmentation performance, which improves until performance saturates around 85% of the training data. It demonstrates that our dataset contains rich and dense supervision, enabling effective training even with fewer subjects.
2. **Challenges of Mixing Incompatible Datasets.** Most public datasets (e.g., BTCV[35]) annotate only a limited number of structures, leading to label granularity mismatch when combined with our 214-class target. Additionally, the partial annotation problem introduces conflicting supervision. For example, structures labeled as “background” in one dataset may be labeled as “foreground” in another, introducing conflicting supervisory signals[36, 37]. While masking background gradients may help, it reduces learning signals and promotes overfitting to the limited labeled regions. Furthermore, sparse annotations lack the anatomical context necessary to model spatial dependencies between neighboring structures, which is a key feature captured by our densely labeled dataset.

We conducted controlled experiments to assess the impact of training volume on model performance. As shown in Fig. 6, the results indicate that segmentation accuracy improves steadily when increasing the dataset size from 30% to 70%, with gains beginning to plateau between 70% and 85%, and reaching near-saturation at 100%. This trend suggests diminishing returns beyond a certain data threshold. We attribute this behavior to the rich inter-organ supervision in our fully annotated, multi-modal dataset.

In conclusion, while we agree that subject diversity is important, we argue that annotation density and multi-organ context are equally crucial for training universal models. We also revised the manuscript to clearly articulate this distinction and explain why incorporating additional public data may compromise training stability.

Comment 2.3: In section 2.3, the authors show good application on using the universal segmentation tech for identifying association ROIs, pairs. From the description, after quantifying the ROIs, the identification of association pairs still needs massive manual efforts. Can this process be automated? Or use as ground truth for training the model, which can help segmentation tasks as well as pre-quantify the regions.

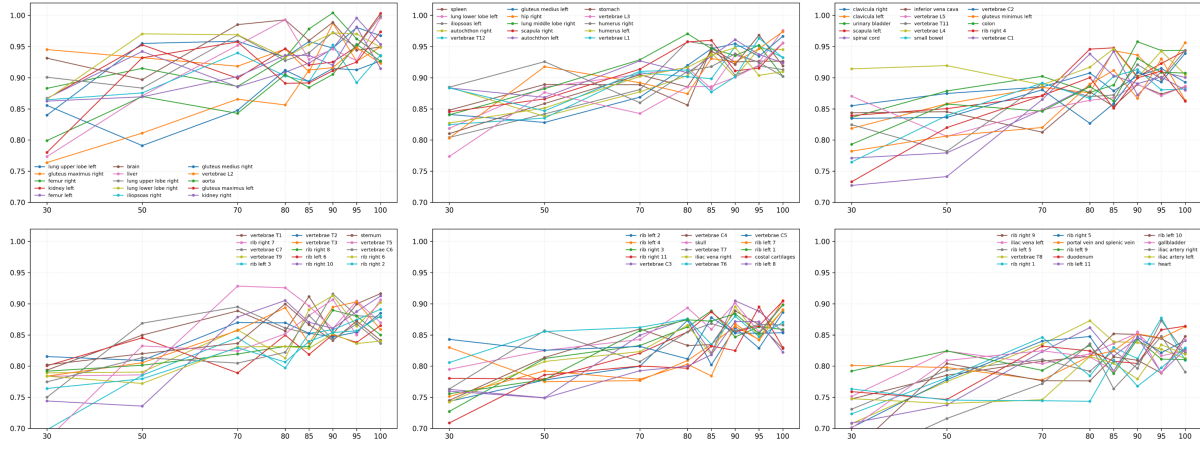


Figure 6: Model performance across varying training data volumes. Each subplot shows the Dice scores for 15 organ categories under different proportions of training data (x-axis: percentage of data used; y-axis: Dice coefficient).

Reply: We appreciate the reviewer’s concern and apologize for the lack of clarity in our original description. The identification of association pairs in our study is fully automated, with no manual annotation or selection is involved. The workflow proceeds as follows:

1. **Automatic ROI segmentation.** The MPUM first segments 215 anatomical regions per scan.
2. **Exhaustive pair enumeration.** A post-processing script then enumerates every unique ROI pairs (20,503 combinations: 3,403 brain–brain, 9,960 brain–body, 7,140 body–body pairs).
3. **Fully automated statistics.** For each pair, the pipeline compute Pearson correlation coefficients, apply Fisher Z-transformation, and derive two-tailed p-values for statistical assessment.

The full pipeline is reproducible and scalable. We will release the code, along with a GUI version, after internal testing and documentation.

Comment 2.4: The body-brain pair for analyzing metabolic associations is a good effort. Is there any association or discovery of any disease-related biomarkers? Such as diabetes/Alzheimer? This part can be extended with more description.

Reply: We thank the reviewer for this constructive suggestion. We address the comment in two parts: (1) by explaining how our framework identifies data-driven metabolic associations as potential imaging biomarkers, and (2) by presenting results from a new Alzheimer’s disease (AD) experiment to demonstrates disease relevance.

In biomarker discovery, imaging features that show statistically significant group differences are often considered candidate biomarkers[38]. Since image-derived characteristic reflecting an underlying biological process, the significant changes we observe in brain–brain or brain–body metabolic associations meet the criteria for candidate imaging biomarkers, such as the coupling between the pallidum and vertebrae T1–T12. These findings, derived through a high-throughput, data-driven whole-body analysis, exemplify how large-scale metabolic profiling can reveal potentially meaningful biomarkers. This is also consistent with regulatory guidelines[39], which define imaging findings with consistent, biologically plausible group-

a**R-Middle and inferior temporal gyrus**

BrainRegion	Corr	P-value	FDR P-value
L-Anterior-temporal-lobe-lateral-part	0.9050 >>> 0.2193	0.0e+00	7.0e-09
R-Medial orbital gyrus	0.8264 >>> 0.0382	2.2e-16	7.0e-09
R-Anterior-temporal-lobe-lateral-part	0.8942 >>> 0.3640	1.7e-14	7.0e-09
R-Fusiform gyrus	0.8429 >>> 0.1911	6.0e-14	7.0e-09
R-Superior frontal gyrus	-0.1828 >>> 0.6413	8.1e-12	7.0e-09
L-Postcentral gyrus	-0.2597 >>> 0.5770	2.4e-11	7.0e-09
R-Postcentral gyrus	-0.1746 >>> 0.6302	3.1e-11	7.0e-09
R-Posterior orbital gyrus	0.8751 >>> 0.4217	6.0e-11	7.9e-09
L-Superior-temporal-gyrus-posterior-part	-0.1640 >>> 0.6074	3.1e-10	1.7e-08
L-Parahippocampal-and-ambient-gyri	0.7712 >>> 0.1719	7.9e-10	3.3e-08
R-Inferiolateral remainder of parietal lobe	0.2962 >>> 0.8135	1.8e-09	6.4e-08
L-Superior parietal gyrus	-0.1406 >>> 0.5814	5.5e-09	1.7e-07
R-Superior parietal gyrus	-0.0061 >>> 0.6596	7.7e-09	2.2e-07
L-Thalamus	-0.3144 >>> 0.4190	2.4e-08	5.7e-07
L-Superior frontal gyrus	-0.1627 >>> 0.5375	3.2e-08	7.3e-07
L-Middle frontal gyrus	0.0449 >>> 0.6678	3.6e-08	8.1e-07
L-Putamen	0.7010 >>> 0.1239	7.2e-08	1.5e-06
R-Parahippocampal-and-ambient-gyri	0.7872 >>> 0.3217	1.3e-07	2.4e-06
R-Pre-subgenual frontal cortex	0.7702 >>> 0.2941	2.1e-07	3.7e-06
L-Medial orbital gyrus	0.8515 >>> 0.5081	3.9e-07	6.2e-06
R-Lateral remainder of occipital lobe	0.1279 >>> 0.6789	4.4e-07	6.6e-06
R-Thalamus	-0.0558 >>> 0.5545	8.5e-07	1.2e-05
L-Inferiolateral remainder of parietal lobe	0.0399 >>> 0.6162	9.1e-07	1.2e-05
R-Caudate nucleus	-0.3173 >>> 0.3365	9.1e-07	1.2e-05
R-Cingulate gyrus gyrus cinguli posterior part	0.7515 >>> 0.3376	6.2e-06	6.2e-05
R-Lateral orbital gyrus	0.8680 >>> 0.6155	1.1e-05	1.0e-04
R-Anterior orbital gyrus	0.8559 >>> 0.5968	2.0e-05	1.6e-04
R-Lingual gyrus	0.0883 >>> 0.5894	2.1e-05	1.7e-04
R-Insula	0.2830 >>> 0.6976	3.5e-05	2.7e-04
L-Precentral gyrus	-0.1895 >>> 0.3595	4.0e-05	3.0e-04
L-Inferior frontal gyrus	0.1999 >>> 0.6470	4.0e-05	3.0e-04
L-Lateral orbital gyrus	0.7890 >>> 0.4633	4.1e-05	3.0e-04
L-Posterior orbital gyrus	0.8302 >>> 0.5550	4.6e-05	3.3e-04
R-Inferior frontal gyrus	0.3527 >>> 0.7299	5.1e-05	3.6e-04
L-Pre-subgenual frontal cortex	0.8177 >>> 0.5447	9.7e-05	6.3e-04
R-Precentral gyrus	0.0074 >>> 0.4907	1.3e-04	8.0e-04
L-Cuneus	0.7330 >>> 0.3960	1.9e-04	1.1e-03
L-Middle and inferior temporal gyrus	0.8494 >>> 0.6353	2.7e-04	1.5e-03
R-Posterior temporal lobe	0.5581 >>> 0.8105	3.1e-04	1.7e-03
R-Hippocampus	0.6495 >>> 0.2843	4.9e-04	2.5e-03
L-Lingual gyrus	0.0306 >>> 0.4314	1.8e-03	7.8e-03
L-Subcallosal area	0.5804 >>> 0.2328	2.1e-03	8.8e-03

b**L-Superior-temporal-gyrus-posterior-part**

BrainRegion	Corr	P-value	FDR P-value
L-Insula	-0.0155 >>> 0.8181	0.0e+00	7.0e-09
L-Cuneus	-0.0691 >>> 0.7171	2.2e-12	7.0e-09
R-Middle and inferior temporal gyrus	-0.1640 >>> 0.6074	3.1e-10	1.7e-08
L-Pre-subgenual frontal cortex	-0.2525 >>> 0.5288	9.1e-10	3.7e-08
L-Posterior temporal lobe	0.3475 >>> 0.8192	1.0e-08	2.8e-07
R-Superior-temporal-gyrus-posterior-part	0.3084 >>> 0.7946	3.1e-08	7.3e-07
R-Putamen	-0.4408 >>> 0.2775	4.2e-08	9.1e-07
L-Straight gyrus	-0.1095 >>> 0.5420	2.1e-07	3.9e-06
L-Postcentral gyrus	0.9012 >>> 0.6517	4.1e-07	6.4e-06
L-Posterior orbital gyrus	-0.2993 >>> 0.3711	4.4e-07	6.6e-06
R-Precentral gyrus	0.8533 >>> 0.5209	5.9e-07	8.7e-06
L-Lateral orbital gyrus	-0.1383 >>> 0.4933	8.8e-07	1.2e-05
R-Posterior orbital gyrus	-0.2480 >>> 0.3987	1.0e-06	1.4e-05
R-Lateral orbital gyrus	-0.1260 >>> 0.4961	1.2e-06	1.6e-05
R-Anterior orbital gyrus	-0.1089 >>> 0.4967	2.2e-06	2.7e-05
R-Straight gyrus	-0.0645 >>> 0.5275	2.5e-06	2.9e-05
R-Lateral ventricle excluding temporal hom	-0.1230 >>> 0.4756	3.6e-06	4.0e-05
L-Lateral ventricle excluding temporal hom	-0.0684 >>> 0.5081	5.4e-06	5.6e-05
L-Medial orbital gyrus	-0.1787 >>> 0.4201	5.5e-06	5.6e-05
L-Middle and inferior temporal gyrus	0.2045 >>> 0.6827	5.8e-06	5.9e-05
R-Parahippocampal-and-ambient-gyri	-0.4870 >>> 0.0775	1.0e-05	9.4e-05
R-Postcentral gyrus	0.8281 >>> 0.5412	3.1e-05	2.4e-04
L-Cingulate gyrus gyrus cinguli anterior part	0.0198 >>> 0.5285	4.0e-05	3.0e-04
R-Cingulate gyrus gyrus cinguli anterior part	0.0066 >>> 0.5135	5.0e-05	3.5e-04
L-Anterior orbital gyrus	0.0936 >>> 0.5667	7.2e-05	4.8e-04
L-Cingulate gyrus gyrus cinguli posterior part	0.8004 >>> 0.5139	1.2e-04	7.6e-04
L-Fusiform gyrus	0.2041 >>> 0.6174	2.0e-04	1.2e-03
L-Subcallosal area	-0.2222 >>> 0.2775	2.2e-04	1.3e-03
R-Superior frontal gyrus	0.8010 >>> 0.5324	2.4e-04	1.4e-03
L-Superior frontal gyrus	0.7977 >>> 0.5344	3.3e-04	1.8e-03
R-Caudate nucleus	0.6911 >>> 0.3704	8.5e-04	4.0e-03
L-Putamen	-0.2309 >>> 0.2055	1.3e-03	6.0e-03
R-Cingulate gyrus gyrus cinguli posterior part	0.0746 >>> 0.4700	1.6e-03	7.1e-03
L-Lateral remainder of occipital lobe	0.4876 >>> 0.7439	2.0e-03	8.7e-03
L-Thalamus	0.7662 >>> 0.5275	2.1e-03	9.0e-03

c**R-Inferiolateral remainder of parietal lobe**

BrainRegion	Corr	P-value	FDR P-value
R-Insula	0.7914 >>> -0.4652	0.0e+00	4.1e-09
R-Middle frontal gyrus	0.9173 >>> -0.3279	0.0e+00	4.1e-09
L-Inferiolateral remainder of parietal lobe	0.8571 >>> -0.4977	0.0e+00	4.1e-09
L-Inferior frontal gyrus	0.6950 >>> -0.6642	0.0e+00	4.1e-09
R-Inferior frontal gyrus	0.8277 >>> -0.5003	0.0e+00	4.1e-09
L-Anterior-temporal-lobe-medial-part	0.3983 >>> -0.7998	2.2e-16	4.1e-09
L-Posterior orbital gyrus	0.5198 >>> -0.7318	2.2e-16	4.1e-09
L-Superior-temporal-gyrus-posterior-part	0.7365 >>> -0.4799	1.3e-15	4.1e-09
R-Anterior orbital gyrus	0.5650 >>> -0.6749	1.8e-15	4.1e-09
R-Superior-temporal-gyrus-posterior-part	0.8604 >>> -0.1436	4.2e-15	4.1e-09
L-Cuneus	0.5559 >>> -0.6656	6.4e-15	4.1e-09
L-Hippocampus	0.4198 >>> -0.7388	2.8e-14	4.1e-09
R-Posterior temporal lobe	0.8252 >>> -0.1792	1.5e-13	4.1e-09
R-Middle and inferior temporal gyrus	0.8135 >>> -0.1907	4.1e-13	4.1e-09
R-Substantia nigra	0.3828 >>> -0.7227	7.1e-13	4.1e-09
R-Precentral gyrus	0.7163 >>> -0.3935	7.2e-13	4.1e-09
L-Substantia nigra	0.3577 >>> -0.7163	3.7e-12	4.1e-09
Third ventricle	0.5030 >>> -0.6063	7.4e-12	4.1e-09
L-Thalamus	0.5842 >>> -0.5203	1.1e-11	4.1e-09
L-Middle and inferior temporal gyrus	0.5357 >>> -0.5619	1.7e-11	4.2e-09
L-Parahippocampal-and-ambient-gyri	0.1372 >>> -0.7867	5.9e-11	5.0e-09
R-Anterior-temporal-lobe-medial-part	0.2669 >>> -0.7048	3.6e-10	1.3e-08
R-Cingulate gyrus gyrus cinguli anterior part	0.7244 >>> -0.2262	4.0e-10	1.4e-08
R-Medial orbital gyrus	0.0279 >>> -0.8029	6.2e-10	1.9e-08
R-Lingual gyrus	0.6714 >>> -0.2840	1.7e-09	4.1e-08
R-Putamen	0.2901 >>> -0.6658	1.9e-09	4.4e-08
L-Straight gyrus	0.6248 >>> -0.2966	1.5e-08	2.6e-07
L-Cingulate gyrus gyrus cinguli anterior part	0.6465 >>> -0.2237	5.5e-08	8.1e-07
L-Anterior orbital gyrus	0.6137 >>> -0.2560	1.0e-07	1.4e-06
L-Lateral ventricle excluding temporal hom	0.5702 >>> -0.3020	1.7e-07	2.2e-06
R-Straight gyrus	0.5917 >>> -0.2626	2.3e-07	2.9e-06
R-Lateral ventricle excluding temporal hom	0.5839 >>> -0.2717	2.4e-07	3.1e-06
R-Hippocampus	0.1536 >>> -0.6548	3.1e-07	3.8e-06
R-Lateral orbital gyrus	0.6073 >>> -0.2080	6.0e-07	6.5e-06
L-Fusiform gyrus	0.6813 >>> -0.0717	8.4e-07	8.8e-06
R-Pre-subgenual frontal cortex	0.1455 >>> -0.6390	8.5e-07	8.9e-06
L-Putamen	-0.0162 >>> -0.7141	1.6e-06	1.6e-05
L-Caudate nucleus	0.3333 >>> -0.4653	3.5e-06	3.1e-05
R-Fusiform gyrus	0.1177 >>> -0.6194	4.4e-06	3.8e-05
R-Precentral gyrus	0.7960 >>> 0.2455	5.0e-06	4.3e-05
R-Posterior orbital gyrus	0.4728 >>> -0.2943	8.4e-06	6.5e-05
L-Posterior temporal lobe	0.7587 >>> 0.1800	9.8e-06	7.4e-05
R-Caudate nucleus	0.3921 >>> -0.3696	1.2e-05	8.9e-05
R-Cerebellum	0.2521 >>> 0.7826	1.5e-05	1.1e-04
L-Medial orbital gyrus	0.3785 >>> -0.3700	1.8e-05	1.2e-04
Corpus callosum	0.3301 >>> -0.3719	6.3e-05	3.8e-04
L-Cingulate gyrus gyrus cinguli posterior part	0.4523 >>> -0.2190	1.1e-04	6.1e-04
R-Anterior-temporal-lobe-lateral-part	0.0673 >>> -0.5657	1.1e-04	6.3e-04
R-Cingulate gyrus gyrus cinguli posterior part	0.4557 >>> -0.1898	1.9e-04	1.0e-03
L-Lateral orbital gyrus	0.4597 >>> -0.1677	2.8e-04	1.4e-03

d**L-Postcentral gyrus**

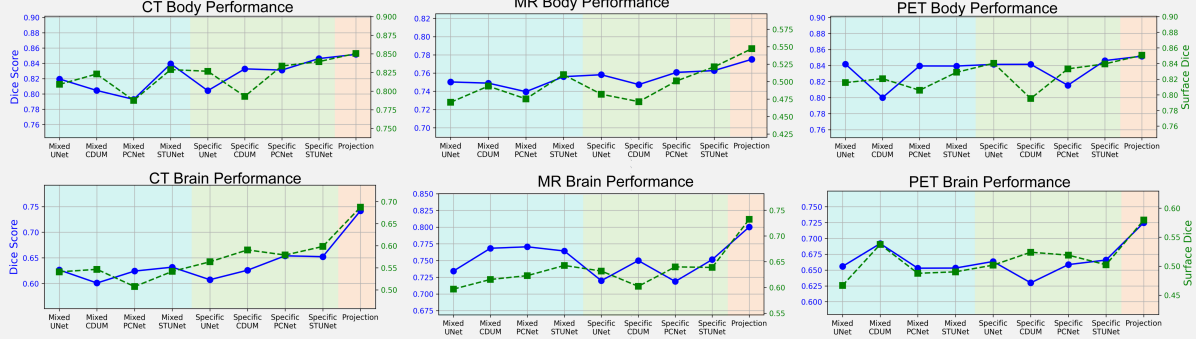
BrainRegion	Corr	P-value	FDR P-value
L-Straight gyrus	0.6265 >>> -0.2864	1.9e-08	3.3e-07
R-Anterior orbital gyrus	0.5240 >>> -0.3882	6.4e-08	9.4e-07
L-Anterior orbital gyrus	0.5336 >>> -0.3385	2.4e-07	3.0e-06
L-Posterior orbital gyrus	0.5457 >>> -0.3203	2.6e-07	3.3e-06
L-Inferiolateral remainder of parietal lobe	0.7207 >>> -0.0157	4.6e-07	5.2e-06
R-Middle frontal gyrus	0.6763 >>> -0.0690	1.2e-06	1.2e-05
R-Precentral gyrus	0.8368 >>> 0.3403	3.1e-06	2.7e-05
L-Inferior frontal gyrus	0.6120 >>> -0.1250	4.9e-06	4.2e-05
L-Cuneus	0.6855 >>> 0.0436	1.4e-05	1.0e-04
R-Cingulate gyrus gyrus cinguli anterior part	0.6658 >>> 0.0109	1.6e-05	1.1e-04
L-Lateral orbital gyrus	0.5139 >>> -0.2149	1.8e-05	1.3e-04
R-Middle and inferior temporal gyrus	0.5770 >>> -0.0809	5.6e-05	3.4e-04
R-Inferior frontal gyrus	0.5398 >>> -0.1185	8.1e-05	4.7e-04
R-Insula	0.6005 >>> 0.0132	2.1e-04	1.1e-03
L-Superior-temporal-gyrus-posterior-part	0.6517 >>> 0.1023	2.3e-04	1.2e-03
L-Thalamus	0.5164 >>> -0.1033	2.3e-04	1.2e-03
L-Medial orbital gyrus	0.2643 >>> -0.3698	3.3e-04	1.6e-03
R-Posterior orbital gyrus	0.3965 >>> -0.2312	3.6e-04	1.7e-03
L-Substantia nigra	0.1859 >>> -0.4355	3.6e-04	1.7e-03
R-Lateral orbital gyrus	0.4659 >>> -0.1428	4.1e-04	1.9e-03
L-Fusiform gyrus	0.4912 >>> -0.1101	4.1e-04	2.0e-03
R-Straight gyrus	0.3760 >>> -0.2403	4.8e-04	2.2e-03
L-Middle frontal gyrus	0.7598 >>> 0.3518	6.2e-04	2.8e-03
R-Substantia nigra	0.1819 >>> -0.4146	6.5e-04	2.9e-03
L-Cingulate gyrus gyrus cinguli anterior part	0.6000 >>> 0.0760	7.7e-04	3.3e-03
R-Posterior temporal lobe	0.4993 >>> -0.0642	8.4e-04	3.6e-03
L-Middle and inferior temporal gyrus	0.3143 >>> -0.2759	9.1e-04	3.8e-03
R-Lingual gyrus	0.7357 >>> 0.3215	9.2e-04	3.9e-03

Figure 7: The impact of Alzheimer’s disease on the metabolic correlations of organs. All results shown are statistically significant at $\alpha = 0.01$ with FDR correction. The “Corr” column in the table is formatted as “A>>>B,” where value A represents the metabolic correlation between two regions under the control group, and value B indicates that in the patient group. Subplots **a** and **b** present the CN vs MCI comparisons, whereas subplots **c** and **d** present the MCI vs AD comparisons.

level differences as valid biomarker candidates.

To further explore disease relevance, we performed an additional analysis using brain 18F-FDG-PET data from the ADNI cohort (97 cognitively normal [CN], 328 mild cognitive impairment [MCI], and 42 Alzheimer’s disease [AD] subjects). Due to the lack of publicly available whole-body PET/CT scans for

a. Aided Identification: Anatomical Structure Identification (case 1)



b. Ablation Experiments

Training Data			CT-Body		CT-Brain	
CT	PET	MR	Dice	Surface Dice	Dice	Surface Dice
✓			0.8125	0.7951	0.6872	0.6071
✓	✓		0.8459	0.8381	0.7223	0.6485
✓		✓	0.8232	0.8116	0.7295	0.6531
✓	✓	✓	0.8517	0.8506	0.7419	0.6872

Figure 8: Comparison of the performance between the MPUM and other advanced models. **a**, Dice Score and surface Dice of CT(left), MR(middle), and PET(right) imaging segmentation tasks. They compare four advanced models: UNet, CDUM, PCNet, and STUNet, with three training strategies: modality-mixed, modality-specific, and modality-projection (used by MPUM). **b**, the table demonstrates the impact of using multi-modality training data on both brain and body tasks in terms of Dice performance.

AD patients, we utilized data from the ADNI, which provides FDG-PET imaging limited to the brain. Following the same pipeline, we included age adjustment and FDR correction. As shown in Fig.7a&b, in the CN vs. MCI comparison, altered associations emerged in the middle and inferior temporal gyri and posterior superior temporal gyrus. These regions are well-known to distinguish MCI or early AD converters from healthy controls[40]. In the MCI vs. AD comparison (Fig. 7 c&d), the most notable changes were found in the inferior lateral parietal cortex and postcentral gyrus. These regions are consistent with prior FDG-PET findings showing parietal hypometabolism as a hallmark of early AD[40, 41]. These results demonstrate the potential of our model to uncover clinically relevant, disease-associated imaging biomarkers.

Our framework is agnostic to disease type and, in principle, can be applied to any condition for which whole-body 18F-FDG PET data are available. At present, the lack of whole-body PET/CT data from diabetic cohorts prevents us from extending our metabolic association analysis to diabetes-related systemic alterations. We view this represents an important direction for future research.

Comment 2.5: From the technical aspect, this paper lacks a general comparison of the proposed universal model; it needs to justify the effectiveness of the multi-modal and training strategy.

Reply: We thank the reviewer for this insightful comment. In Case 1, we directly quantified the effectiveness of multi-modal and training strategy. Besides, in the revised manuscript we have added a new set of experiments that compare MPUM against three universal segmentation models, including TotalSegmentator[33], MedSAM[28] and Vista3D[32].

Table 2: Dice coefficient comparison across organs. Bold numbers denote the best-performing model for each organ. † indicates a statistically significant difference ($p < 0.05$) from TotalSegmentator, and ‡ from Vista3D. * indicates a statistically significant difference from MPUM ($p < 0.05$).

Organ	TotalSeg.	Vista3D	MPUM	Organ	TotalSeg.	Vista3D	MPUM
adrenal gland right	0.785	0.743	0.785 †	aorta	0.943	0.921	0.945 †
autochthon left	0.950	0.928	0.944 †	autochthon right	0.944	0.923	0.945 †
brain	0.985	0.924	0.979 †	clavicula left	0.943 *	0.860	0.928 †
clavicula right	0.933	0.867	0.929 †	colon	0.861	0.834	0.904 †
costal cartilages	0.761	0.685	0.855 † ‡	duodenum	0.826	0.804	0.832 †
femur left	0.960	0.920	0.966 †	femur right	0.971	0.906	0.966 †
gallbladder	0.824 *	0.790	0.816 †	gluteus maximus left	0.939	0.920	0.951 † ‡
gluteus maximus right	0.938	0.910	0.949 † ‡	gluteus medius left	0.921	0.910	0.946 † ‡
gluteus medius right	0.919	0.907	0.942 † ‡	gluteus minimus left	0.851	0.837	0.903 † ‡
heart	0.378	0.365	0.816 † ‡	hip right	0.960	0.922	0.953 †
humerus left	0.901	0.635	0.937 † ‡	humerus right	0.904	0.699	0.936 † ‡
iliac artery left	0.811	0.775	0.819 †	iliac artery right	0.830	0.766	0.831 †
iliac vena left	0.859	0.831	0.851	iliac vena right	0.838	0.814	0.858 † ‡
iliopsoas left	0.935	0.924	0.935 †	iliopsoas right	0.939	0.926	0.944 † ‡
inferior vena cava	0.893	0.872	0.903 †	kidney left	0.930	0.923	0.955 † ‡
kidney right	0.936	0.935	0.951 †	liver	0.969	0.940	0.971 †
lung lower lobe left	0.968	0.929	0.952 †	lung lower lobe right	0.957	0.942	0.967
lung middle lobe right	0.936	0.876	0.942 †	lung upper lobe left	0.972	0.944	0.975
lung upper lobe right	0.965	0.946	0.969 †	portal vein/splenic vein	0.849	0.789	0.844 †
spinal cord	0.765	0.744	0.914 † ‡	rib left 1	0.857	0.766	0.862 † ‡
rib left 2	0.777	0.725	0.863 † ‡	rib left 3	0.786	0.700	0.869 † ‡
rib left 4	0.793	0.704	0.867 † ‡	rib left 5	0.817	0.707	0.867 † ‡
rib left 6	0.821	0.724	0.870 † ‡	rib left 7	0.825	0.734	0.862 † ‡
rib left 8	0.821	0.708	0.858 † ‡	rib left 9	0.839	0.722	0.851 †
rib left 10	0.861 *	0.773	0.830 † ‡	rib left 11	0.812	0.779	0.837 † ‡
rib right 1	0.861	0.767	0.847 † ‡	rib right 2	0.877 *	0.771	0.861 †
rib right 3	0.833	0.726	0.873 † ‡	rib right 4	0.852	0.806	0.893 † ‡
rib right 5	0.672	0.875	0.860 †	rib right 6	0.813	0.741	0.882 † ‡
rib right 7	0.804	0.737	0.880 † ‡	rib right 8	0.800	0.705	0.878 † ‡
rib right 9	0.785	0.703	0.874 † ‡	rib right 10	0.780	0.679	0.860 † ‡
rib right 11	0.763	0.713	0.866 † ‡	spleen	0.955	0.948	0.948
scapula left	0.911	0.832	0.925 † ‡	scapula right	0.912	0.843	0.926 † ‡
skull	0.787	0.740	0.884 † ‡	small bowel	0.867	0.836	0.908 † ‡
sternum	0.572	0.539	0.886 † ‡	stomach	0.923	0.905	0.939 †
urinary bladder	0.904	0.878	0.927 † ‡	vertebrae C1	0.863	0.806	0.890 † ‡
vertebrae C2	0.923 *	0.857	0.909 †	vertebrae C3	0.895	0.836	0.888 †
vertebrae C4	0.882	0.829	0.883 †	vertebrae C5	0.876	0.813	0.877 †
vertebrae C6	0.879	0.806	0.877 †	vertebrae C7	0.899	0.831	0.893 †
vertebrae L1	0.919	0.895	0.943 † ‡	vertebrae L2	0.919	0.897	0.934 † ‡
vertebrae L3	0.920	0.889	0.920 †	vertebrae L4	0.921	0.867	0.926 †
vertebrae L5	0.921	0.888	0.921 †	vertebrae T1	0.909	0.847	0.904 †
vertebrae T11	0.937 *	0.892	0.919 † ‡	vertebrae T12	0.918	0.895	0.941 † ‡
vertebrae T2	0.901	0.847	0.897 †	vertebrae T3	0.885	0.839	0.888 †
vertebrae T5	0.879	0.790	0.885 †	vertebrae T6	0.878	0.809	0.887 †
vertebrae T7	0.886	0.823	0.882 †	vertebrae T8	0.903 *	0.841	0.862
vertebrae T9	0.916 *	0.867	0.894 †				

Effectiveness of Training Strategies. In Fig. 8, we compare three multi-modality training strategies: (1) modality-mixed, where a single model is trained on all modalities with shared parameters; (2) modality-specific, where separate models are trained per modality; and (3) our proposed modality-projection strategy, where shared latent representation is dynamically projected into modality-specific convolutional kernels via a controller module. While the modality-mixed approach benefits from data scale, it suffers from conflicting optimization gradients across modalities. The modality-specific strategy avoids such conflict but misses the benefits of cross-modal learning. In contrast, our projection method resolves this tension by learning a shared latent representation and dynamically projecting it into modality-specific convolutional kernels using a controller module. As shown in Fig. 8a, this approach achieves the highest Dice and surface-Dice scores across all benchmarks (e.g., MRI body: 0.7751 vs. 0.7560/0.7627; CT brain: 0.7419 vs. 0.6318/0.6540), confirming its robustness and effectiveness.

Effectiveness of multi-modal. As shown in Fig. 8b, we conducted a multi-modal ablation study by training identical segmentation models under four input settings (CT only, CT+PET, CT+MR, and CT+PET+MR) and evaluated both body and brain tasks. Compared to the CT-only baseline, adding PET or MR each yielded clear improvements in Dice and surface-Dice, reflecting complementary anatomical and functional cues. Finally, the tri-modal model achieved the best performance overall, confirming that integrating multiple modalities confers synergistic improvements in segmentation accuracy.

Comparative Evaluation with Universal Models. We further benchmarked MPUM against two universal segmentation frameworks (TotalSegmentator and Vista3D) on a single organ analysis in Tab. 2. MPUM attains the highest Dice in the majority of the categories, with statistically significant gains ($p < 0.05$) in over 60 organs. It matches or slightly outperforms existing models on high-contrast targets like the liver and spleen, while showing notable advantages in challenging structures such as vessels (e.g., inferior vena cava: 0.903 vs. 0.893/0.872), musculoskeletal regions (e.g., gluteus maximus: 0.951 vs. 0.939/0.920), and small abdominal organs (e.g., colon: 0.904 vs. 0.861/0.834). As for interactive segmentation frameworks like MedSAM[28], we have already discussed their limitations in Reply to Comment 2.1. Briefly, their dependence on manual prompts (e.g., points or bounding boxes) renders them unsuitable for fully automated, high-throughput applications, such as the use cases addressed by our proposed MPUM.

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On Revision of NCOMMS-24-81749

“Modality-Projection Universal Model for Comprehensive Full-Body Medical Imaging Segmentation”

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Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han,
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August 9, 2025

Dear Editor and Reviewers,

Thank you very much for your valuable feedback and suggestions on our manuscript [Manuscript ID: NCOMMS-24-81749]. We greatly appreciate the time and effort you have invested in reviewing our work, and we are grateful for the constructive comments that have helped us improve the quality of our manuscript. We have carefully considered all the comments and have made revisions to address the concerns raised.

Reply to Reviewer #2’s Comments

Comment 1.1: Thank you for addressing the points raised. The manuscript has clearly improved as a consequence.

Reply: We thank the reviewer for recognizing the improvements made in our first-round revision.

Reply to Reviewer #3’s Comments

Comment 2.1: As I have commented in the first round review, this paper has a unique impact on how to do translational research from a pioneering deep learning model to clinical settings. The authors can further add the discussion of the value of AI’s impact on real clinical workflow, and highlight how to bridge frontier tech with medicine.

Reply: We thank the reviewer for the insightful comment.

Here we envision MPUM as a reusable imaging module within broader clinical AI pipelines. Its structured outputs, including whole-body and brain ROI labels, volumetric measurements, and metabolic matrices, serve as standardized inputs for a wide spectrum of downstream tasks. These span rapid localization to support acute clinical workflows; standardized feature extraction for population-scale imaging studies, enabling cross-organ and cross-modality analysis; and automated reporting or decision-support systems, where MPUM outputs feed directly into multimodal analytics and domain-specific large language models (LLMs). To translate this potential from prototype to bedside, future work will focus

on addressing key technical challenges such as domain shift, continual learning, and secure deployment. We also aim to expand multi-task learning and integrate MPUM seamlessly into hospital infrastructure, ultimately enabling a stepwise transition toward AI-assisted, imaging-driven clinical care.